



Cordian B. Riener

Curriculum vitae – September 2026

Personal information

First name:	Cordian Benedikt	Surname:	Riener
Sex:	Male	Website:	https://www.cordian.de
Date of birth:	April 9 th 1981	ORCID:	0000-0002-1192-3500
Nationality:	Austrian, German		

Research profile

My research lies at the interface of real algebraic geometry, polynomial optimization and symbolic computation, and is driven by one central question: how can symmetry be turned into a computational resource? Deciding whether a polynomial is nonnegative, describing the topology of a semi-algebraic set, solving polynomial systems or certifying bounds in discrete geometry are computationally hard in general, but the instances that matter in practice almost always carry symmetry. I develop the mathematical theory that makes this symmetry usable: degree principles that reduce symmetric problems in n variables to a bounded number of variables, equivariant bounds and algorithms for the topology of symmetric semi-algebraic sets (with S. Basu), symmetry-reduced semidefinite relaxations and quadrature rules with few nodes (with M. Schweighofer), and effective descriptions of orbit spaces and symmetric ideals for finite reflection groups. A recurring theme is the interplay between structure and complexity: the same symmetry that makes problems tractable in fixed degree leads, as the number of variables grows, to sharp hardness and undecidability results – for instance for trace inequalities of symmetric matrices, via the geometry of the Vandermonde map. This work connects to extremal combinatorics, discrete geometry and the foundations of computation, and forms the core of the research programme “The Computational Power of Symmetry”. Current directions extend these ideas to equivariant machine learning and the structure of neural networks.

Education

2018 **Habilitation in Mathematics**, *Universität Konstanz*, Germany
2013 **Magister Artium in Philosophy**, *Goethe University Frankfurt*, Germany
2011 **Doctorate (Dr. phil. nat.) in Mathematics**, *Goethe University Frankfurt*, Germany
2007 **Bachelor of Arts in Philosophy**, *University of Ulm*, Germany
2006 **Diplom in Mathematical Economics**, *University of Ulm*, Germany
2005 **Master 2 in Pure Mathematics**, *Université de Bordeaux 1*, France

Honours and academy memberships

2026– **Elected Member**, *The Royal Norwegian Society of Sciences and Letters*, Det Kongelige Norske Videnskabers Selskab (DKNVS), Trondheim

Department of Mathematics and Statistics, UiT – The Arctic University of Norway
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Employment History

- 2021–Present **Vice Dean for PhD Education (40% position)**, *Faculty of Science and Technology, UiT – The Arctic University of Norway*
- 2019–Present **Professor of Mathematics**, *UiT – The Arctic University of Norway*
- 2018–2019 **Associate Professor of Mathematics**, *UiT – The Arctic University of Norway*
- 2016–2018 **Interim Professor**, *Universität Konstanz, Germany*
- 2012–2016 **Aalto Science Fellow**, *Aalto University, Finland*
- 2011–2012 **Postdoctoral Fellow**, *Zukunftskolleg, Universität Konstanz, Germany*
- 2007–2011 **Doctoral Researcher**, *Goethe University Frankfurt, Germany*

Institutional and Academic Responsibilities

Leadership positions

- 2025– **Scientific leader**, *Mathesis – PhD School for Foundations of Computational Mathematics*, national PhD school
- 2024– **Co-Director**, *Lie-Størmer Center*, national research center for mathematics, UiT/University of Bergen
- 2023– **Chair**, *Norwegian Mathematics Council*
- 2020 **Acting Vice Head**, *Department of Mathematics and Statistics*, UiT – The Arctic University of Norway
- 2019– **Leader**, *Applied and Computational Algebra Research Group*, UiT – The Arctic University of Norway

Editorial activities

- 2026– **Editor-in-Chief**, *Journal of Symbolic Computation*
- Series Editor**, *Fundamental Structures in Computational and Pure Mathematics*, Birkhäuser/Springer book series (subseries of *Trends in Mathematics*), with K. Ebrahimi-Fard and H. Munthe-Kaas
- 2024–2025 **Editor**, *Journal of Symbolic Computation*
- 2023 **Editor**, *Polynomial Optimization, Moments, and Applications*, Springer Optimization and Its Applications, vol. 206, with M. Kočvara and B. Mourrain
- 2020 **Guest Editor**, *Journal of Pure and Applied Algebra*, special issue *Arctic Applied Algebra*, with S. Basu and M.-F. Roy

Boards, committees and panels

- 2025– **Member**, *SURE-AI – Norwegian national research center for AI*, UiT partner institution
- 2028 **Member**, *NSERC Discovery Grants Evaluation Group, Mathematics and Statistics*, Canada
- Member**, *Scientific Advisory Board, Heidelberg Laureate Forum*
- Member**, *Internal Policy Group, European Mathematical Society*
- 2022– **Board Member**, *Abel Prize*, Norwegian Academy of Science and Letters
- 2022, 2023 **Secretary**, *SIAM Activity Group on Algebraic Geometry*
- 2021 **Committee Member**, *SIAG/Algebraic Geometry Early Career Prize*, SIAM
- 2020– **Scientific Advisory Board Member**, *Effective Methods in Algebraic Geometry (MEGA)*
- 2020– **Steering Board Member**, *Norwegian Mathematical Society*
- 2019–2020 **Faculty Board Member**, *Faculty of Science and Technology*, UiT – The Arctic University of Norway
- 2014–2015 **Steering Group Member**, *Aalto Science Institute*, Aalto University, Finland

2009–2011 **Faculty Board Member**, *Faculty of Mathematics and Computer Science*, Goethe University Frankfurt, Germany

Research funding and project leadership

Overall volume of the projects listed below, as PI, co-PI or local PI: approximately 95 MNOK ($\approx$ €8M), counting the local shares of the EU networks.

- 2025–2031 **Co-Director**, *Lie–Størmer Center*, Budget: $\sim$ 25.5 MNOK
- 2025–2032 **Scientific Leader**, *Mathesis – PhD School for Foundations of Computational Mathematics*, Budget: $\sim$ 16 MNOK
- 2025–2028 **National scientific leader**, *Mathematical Theory of Data Transmission and Data Encryption*, Indo-Norwegian Cooperation Programme in Higher Education and Research (INCP2-2024/10213), Budget: $\sim$ 2.8 MNOK
- 2024–2027 **Local Principal Investigator**, *Tensor Modeling, Geometry and Optimization (TENORS)*, HORIZON–MSCA-2023-DN-JD, Total budget: $\sim$ €4.5M, local share: $\sim$ €600K
- 2022–2025 **Co-Principal Investigator**, *Mathematical Structures in Computations*, Aurora Initiative, UiT Strategic University Fund, Budget: $\sim$ 30 MNOK
- 2022–2025 **Principal Investigator**, *Symmetry in Quantitative and Algorithmic Real Algebraic Geometry (Part 2)*, Trond Mohn Foundation, Budget: $\sim$ 4 MNOK
- 2019–2022 **Local Principal Investigator**, *Polynomial Optimization: Efficiency Through Moments and Algebra (POEMA)*, MSCA-ITN, Total budget: $\sim$ €4M, local share: $\sim$ €300K
- 2018–2022 **Principal Investigator**, *Symmetry in Quantitative and Algorithmic Real Algebraic Geometry (Part 1)*, Trond Mohn Foundation, Budget: $\sim$ 7 MNOK

Supervision and mentoring

Master's students	Ph.D. students	University	Country
1		Goethe University Frankfurt	Germany
6		Universität Konstanz	Germany [‡]
1		FU Berlin	Germany*
1	1	University of Cologne	Germany*
1		Universität Wien	Austria
5	7	UiT – The Arctic University of Norway	Norway [†]

* joint co-supervision ‡ one thesis jointly co-supervised † including current PhD students

Master's 15 theses supervised; five graduates went on to doctoral studies.

PhD students Four completed (Jan Rolfes, now Associate Professor at Linköping University; Sebastian Debus and Robin Schabert, both now at the University of Konstanz), four ongoing.

Postdocs Two current Marie Skłodowska-Curie Fellows (Tobias Boege, Daniel Brosch; applications supported by me), four former; two of them now *maîtres de conférences* (Philippe Moustrou, Toulouse III; Thi Xuan Vu, Lille).

Teaching

At UiT I teach across the whole curriculum, from first-year linear algebra and calculus to algebra, real algebraic geometry and numerical methods, complemented by special curricula on polynomial optimization, invariant and representation theory, and symmetry in computation. Since 2021 I have organized nationwide courses for master's students from universities across Norway (weekly online lectures combined with in-person meetings in Tromsø and Bergen), now run through the Lie–Størmer Center. In 2026 I launched two online micro-courses for a broad audience, *What Is*

Mobility: visits of duration at least three months

- 2015 Fields Institute: Thematic Program on Computer Algebra
- 2010 IPAM (UCLA): Program on Modern Trends in Optimization and Its Application

Selected invited lectures

- 2026 Academy lecture, *Nye medlemmer presenterer*, The Royal Norwegian Society of Sciences and Letters (DKNVS), Trondheim.
- 2025 Invited tutorial, **ISSAC 2025** (50th International Symposium on Symbolic and Algebraic Computation), CIMAT, Guanajuato, Mexico.
- 2025 Invited lecture, Nordic Number Theory Network (N^3) meeting, Stockholm.
- 2024 Semi-plenary lecture, Norwegian Meeting of Mathematicians, Oslo.
- 2023 Invited lecture, **FoCM 2023** (Foundations of Computational Mathematics), workshop *Real-Number Complexity*, Paris.
- 2023 Colloquium lecture, Department of Mathematics and Statistics, Universität Konstanz.
- 2023 Invited lecture, Oberwolfach workshop *New Directions in Real Algebraic Geometry*, MFO.
- 2018 Invited lecture, CIRM conference *Symmetry and Computation*, Luminy.
- 2015 Invited lecture, Fields Institute Thematic Program on Computer Algebra, Toronto (*Bounds on equivariant Betti numbers of symmetric semi-algebraic sets*).
- 2014 Invited lecture, IPAM (UCLA), long program *Algebraic Techniques for Combinatorial and Computational Geometry*, Los Angeles.
- 2013–2023 Invited minisymposium talks at the biennial SIAM Conferences on Applied Algebraic Geometry.

Organization of Scientific Events (Selection)

- 2027 Main organizer, *Polynomials in Tromsø* (PIT 2027), UiT Tromsø.
- 2026 Co-organizer, NCM research school on real algebraic geometry and workshop *Mathematical Structures*, IIT Bombay, India.
- 2025 Co-organizer, MFO-RIMS Tandem Workshop: *Optimization, Theoretical Computer Science and Algebraic Geometry: Convexity and Beyond*.
- 2023 Chair, local organization of **ISSAC** (International Symposium on Symbolic and Algebraic Computation), Tromsø.
- 2023 Co-organizer, CIRM Workshop: *Symmetry, Stability, and Interactions with Computation*.
- 2022 Co-organizer, Norwegian National Mathematicians Meeting, Tromsø.
- 2022 Co-organizer, 14th Nordic Combinatorial Conference, Tromsø.
- 2019–2022 Main organizer, annual Mørketidens Mattemøte, UiT Tromsø*.
- 2021 Co-organizer, MFO-RIMS Tandem Workshop: *Symmetries on Polynomial Ideals and Varieties*.
- 2021 Chair, local organization of **MEGA** (Effective Methods in Algebraic Geometry) Conference.
- 2020 Co-organizer, ICERM Workshop: *Symmetry, Randomness and Computations*.
- 2019 Chair, local organization of the Arctic Applied Algebra Conference.
- 2018 Main organizer, Summer School on Algebraic Statistics, UiT Tromsø.
- 2018 Session organizer, 23rd International Symposium on Mathematical Programming.
- 2014 Session organizer, IFORS Conference, Barcelona.
- 2014 Co-organizer, *Applications of Real Algebraic Geometry*, Aalto University.
- 2013–2023 Organizer, mini-symposia at the biennial SIAM Conferences on Applied Algebraic Geometry.
- 2012 Co-organizer, Workshop on *Mathematics, Algorithms and Proof*, University of Konstanz.

* Annual meeting for Scandinavian mathematicians in Tromsø during the "Mørketid" – the dark season from November to January.

Outreach activities

I am passionate about sharing my fascination for mathematics also outside the normal scientific channels. I developed the *Arctic Mathematical Talent Initiative* (AMTI), a programme that brings gifted upper secondary students to the university, where they complete the highest school mathematics level a year early and then take dedicated university courses with close mentoring by students; AMTI forms the first stage of a three-tier talent programme (school – university – PhD) at the Lie–Størmer Center. I am actively involved in UiT Skolelab, where school classes can visit the university, and the Northern Norwegian science center, where I organized events for visitors to explore mathematics. Furthermore, I initiated the LAUMA (**L**earning and **U**nderstanding **M**athematics) initiative which aims to focus outreach activities and to cross academic boundaries to discuss, explore, and research the status of mathematics as part of our society, and I organized events to promote the diversity of mathematics, in particular the Women of Mathematics photo exhibition at UiT. Finally, I write regularly for a general audience: my opinion pieces on mathematics as part of our culture and on the impact of AI on mathematics and science have appeared in Norway's leading national media, including *Aftenposten*, *Klassekampen*, *Morgenbladet* and *Khrono*.

Publications

Five selected publications

- Saugata Basu and Cordian Riener. “Bounding the equivariant Betti numbers of symmetric semi-algebraic sets”. In: *Advances in Mathematics* 305 (2017), pp. 803–855 — The first polynomial bounds on the equivariant Betti numbers of symmetric semi-algebraic sets; this paper opened the quantitative theory of symmetric semi-algebraic sets.
- Grigoriy Blekherman and Cordian Riener. “Symmetric nonnegative forms and sums of squares”. In: *Discrete & Computational Geometry* 65 (2021), pp. 764–799 — A systematic study of the gap between symmetric nonnegative forms and sums of squares, including the limit as the number of variables grows.
- Jose Acevedo, Grigoriy Blekherman, Sebastian Debus, and Cordian Riener. “The wonderful geometry of the Vandermonde map”. In: *Foundations of Computational Mathematics* 26 (2026), pp. 2005–2051 — The geometry of the image of the power-sum map and, as a consequence, undecidability of nonnegativity for trace inequalities of symmetric matrices of all sizes.
- Cordian Riener, Thorsten Theobald, Lina Jansson Andrén, and Jean B. Lasserre. “Exploiting symmetries in SDP-relaxations for polynomial optimization”. In: *Mathematics of Operations Research* 38.1 (2013), pp. 122–141 — A foundational reference for symmetry reduction in the moment-SOS hierarchy of polynomial optimization.
- Cordian Riener and Markus Schweighofer. “Optimization approaches to quadrature: New characterizations of Gaussian quadrature on the line and quadrature with few nodes on plane algebraic curves, on the plane and in higher dimensions”. In: *Journal of Complexity* 45 (2018), pp. 22–54 — Optimization-theoretic characterizations of Gaussian quadrature and the existence of quadrature rules with few nodes on plane algebraic curves and in higher dimensions.

Journal Articles

- [1] Jose Acevedo, Grigoriy Blekherman, Sebastian Debus, and Cordian Riener. “The wonderful geometry of the Vandermonde map”. In: *Foundations of Computational Mathematics* 26 (2026), pp. 2005–2051.
- [2] Sebastian Debus, Cordian Riener, and Robin Schabert. “Slices of stable polynomials and connections to the Grace–Walsh–Szegő theorem”. In: *Journal of Symbolic Computation* 133 (2026), p. 102488.
- [3] István Miklós and Cordian Riener. “#P-hardness proofs of matrix immanants evaluated on restricted matrices”. In: *Theoretical Computer Science* 1062 (2026), p. 115660.
- [4] Philippe Moustrou, Cordian Riener, and Robin Schabert. “Constructively describing orbit spaces of finite groups by few inequalities”. In: *Journal of Symbolic Computation* 132 (2026), p. 102471.
- [5] Cordian Riener, Jan Rolfes, and Frank Vallentin. “A semidefinite programming hierarchy for covering problems in discrete geometry”. In: *Numerical Algebra, Control and Optimization* 16 (2026), pp. 157–170.

- [6] Córdian Riener, Robin Schabert, and Thi Xuan Vu. “Deciding connectivity in symmetric semi-algebraic sets”. In: *Journal of Symbolic Computation* 138 (2026).
- [7] Córdian Riener and Ettore Teixeira Turatti. “Quadrature rules with few nodes supported on algebraic curves”. In: *Journal of Complexity* 96 (2026), p. 102067.
- [8] Jose Acevedo, Grigoriy Blekherman, Sebastian Debus, and Córdian Riener. “Symmetric nonnegative functions, the tropical Vandermonde cell and superdominance of power sums”. In: *Collectanea Mathematica* (2025). Online first.
- [9] Sebastian Debus, Charu Goel, Salma Kuhlmann, and Córdian Riener. “On nonnegative invariant quartics in type A”. In: *Journal of Symbolic Computation* 128 (2025), p. 102393.
- [10] Evelyne Hubert, Tobias Metzlauff, Philippe Moustrou, and Córdian Riener. “Optimization of trigonometric polynomials with crystallographic symmetry and spectral bounds for set avoiding graphs”. In: *Mathematical Programming* 213.1 (2025), pp. 517–573.
- [11] Mario Kummer and Córdian Riener. “Equivariant algebraic and semi-algebraic geometry of infinite affine space”. In: *Journal of Algebra* 666 (2025), pp. 28–46.
- [12] Philippe Moustrou, Córdian Riener, Thorsten Theobald, and Hugues Verdure. “Symmetric SAGE and SONC forms, exactness and quantitative gaps”. In: *Journal of Symbolic Computation* 127 (2025), p. 102374.
- [13] Ragni Piene, Córdian Riener, and Boris Shapiro. “Return of the plane evolute”. In: *Annales de l’Institut Fourier* 75.4 (2025), pp. 1685–1751.
- [14] Evelyne Hubert, Tobias Metzlauff, and Córdian Riener. “Orbit spaces of Weyl groups acting on compact tori: A unified and explicit polynomial description”. In: *SIAM Journal on Applied Algebra and Geometry* 8.3 (2024).
- [15] Córdian Riener and Robin Schabert. “Linear slices of hyperbolic polynomials and positivity of symmetric polynomial functions”. In: *Journal of Pure and Applied Algebra* 228.5 (2024), p. 107552.
- [16] Sebastian Debus, Philippe Moustrou, Córdian Riener, and Hugues Verdure. “The poset of Specht ideals for hyperoctahedral groups”. In: *Algebraic Combinatorics* 6.6 (2023), pp. 1593–1619.
- [17] Sebastian Debus and Córdian Riener. “Reflection groups and cones of sums of squares”. In: *Journal of Symbolic Computation* 119 (2023), pp. 112–144.
- [18] Gabriella Óturai, Córdian Riener, and Sarah E. Martiny. “Attitudes towards mathematics, achievement, and drop-out intentions among STEM and Non-STEM students in Norway”. In: *International Journal of Educational Research Open* 4 (2023), p. 100230.
- [19] Saugata Basu and Córdian Riener. “Vandermonde varieties, mirrored spaces, and the cohomology of symmetric semi-algebraic sets”. In: *Foundations of Computational Mathematics* 22.5 (2022), pp. 1395–1462.
- [20] Evelyne Hubert, Tobias Metzlauff, Philippe Moustrou, and Córdian Riener. “T-orbit spaces of multiplicative actions and applications”. In: *ACM Communications in Computer Algebra* 56.2 (2022), pp. 72–75.
- [21] Philippe Moustrou, Helen Naumann, Córdian Riener, Thorsten Theobald, and Hugues Verdure. “Symmetry reduction in AM/GM-based optimization”. In: *SIAM Journal on Optimization* 32.2 (2022), pp. 765–785.
- [22] Grigoriy Blekherman and Córdian Riener. “Symmetric nonnegative forms and sums of squares”. In: *Discrete & Computational Geometry* 65 (2021), pp. 764–799.
- [23] Philippe Moustrou, Córdian Riener, and Hugues Verdure. “Symmetric ideals, Specht polynomials and solutions to symmetric systems of equations”. In: *Journal of Symbolic Computation* 107 (2021), pp. 106–121.
- [24] Saugata Basu and Córdian Riener. “On the isotypic decomposition of cohomology modules of symmetric semi-algebraic sets: Polynomial bounds on multiplicities”. In: *International Mathematics Research Notices* 2020.7 (2020), pp. 2054–2113.

- [25] Grigoriy Blekherman, Mario Kummer, Cordian Riener, Markus Schweighofer, and Cynthia Vinzant. “Generalized eigenvalue methods for Gaussian quadrature rules”. In: *Annales Henri Lebesgue* 3 (2020), pp. 1327–1341.
- [26] Saugata Basu and Cordian Riener. “On the equivariant Betti numbers of symmetric definable sets: Vanishing, bounds and algorithms”. In: *Selecta Mathematica* 24.4 (2018), pp. 3241–3281.
- [27] Tobias Friedl, Cordian Riener, and Raman Sanyal. “Reflection groups, reflection arrangements, and invariant real varieties”. In: *Proceedings of the American Mathematical Society* 146.3 (2018), pp. 1031–1045.
- [28] Cordian Riener and Markus Schweighofer. “Optimization approaches to quadrature: New characterizations of Gaussian quadrature on the line and quadrature with few nodes on plane algebraic curves, on the plane and in higher dimensions”. In: *Journal of Complexity* 45 (2018), pp. 22–54.
- [29] Saugata Basu and Cordian Riener. “Bounding the equivariant Betti numbers of symmetric semi-algebraic sets”. In: *Advances in Mathematics* 305 (2017), pp. 803–855.
- [30] Cordian Riener and Nicolai Vorobjov. “On irreducible components of real exponential hypersurfaces”. In: *Arnold Mathematical Journal* 3.3 (2017), pp. 423–443.
- [31] Paul Görlach, Cordian Riener, and Tillmann Weißer. “Deciding positivity of multisymmetric polynomials”. In: *Journal of Symbolic Computation* 74 (2016), pp. 603–616.
- [32] Cordian Riener. “Symmetric semi-algebraic sets and non-negativity of symmetric polynomials”. In: *Journal of Pure and Applied Algebra* 220.8 (2016), pp. 2809–2815.
- [33] Cordian Riener, Thorsten Theobald, Lina Jansson Andrén, and Jean B. Lasserre. “Exploiting symmetries in SDP-relaxations for polynomial optimization”. In: *Mathematics of Operations Research* 38.1 (2013), pp. 122–141.
- [34] Alexander Kovačec, Salma Kuhlmann, and Cordian Riener. “A note on extrema of linear combinations of elementary symmetric functions”. In: *Linear and Multilinear Algebra* 60.2 (2012), pp. 219–224.
- [35] Cordian Riener. “On the degree and half-degree principle for symmetric polynomials”. In: *Journal of Pure and Applied Algebra* 216.4 (2012), pp. 850–856.
- [36] Cordian Riener and Thorsten Theobald. “Positive Polynome und semidefinite Optimierung”. In: *Jahresbericht der Deutschen Mathematiker-Vereinigung* 110.2 (2008), pp. 57–76.
- [37] Cordian Riener. “On extreme forms in dimension 8”. In: *Journal de Théorie des Nombres de Bordeaux* 18.3 (2006), pp. 677–682.

Preprints

- [1] Jose Acevedo, Grigoriy Blekherman, Sebastian Debus, Seokbin Lee, and Cordian Riener. *Eigenvalues of locally positive semidefinite matrices: Non-convexity and geometry*. 2026. arXiv: 2608.15444.
- [2] Henri Breloer and Cordian Riener. *Symmetric and isotypic Hilbert series for symmetric ideals*. 2026. arXiv: 2606.18895.
- [3] Cordian Riener and Thi Xuan Vu. *Symbolic Computation with Symmetric Polynomials in Real Algebraic Geometry*. 2025. arXiv: 2507.23728.

Conference Proceedings

- [1] Cordian Riener, Robin Schabert, and Thi Xuan Vu. “Connectivity in Symmetric Semi-Algebraic Sets”. In: *Proceedings of the 2024 International Symposium on Symbolic and Algebraic Computation*. ISSAC '24. Conference version. Association for Computing Machinery, 2024, pp. 162–169.
- [2] George Labahn, Cordian Riener, Mohab Safey El Din, Éric Schost, and Thi Xuan Vu. “Faster real root decision algorithm for symmetric polynomials”. In: *Proceedings of the 2023 International Symposium on Symbolic and Algebraic Computation*. ISSAC '23. Association for Computing Machinery, 2023, pp. 452–460.

- [3] Cordian Riener and Mohab Safey El Din. “Real root finding for equivariant semi-algebraic systems”. In: *International Symposium on Symbolic and Algebraic Computation*. ISSAC '18. Conference version. Association for Computing Machinery, 2018, pp. 335–342.

Book Chapters

- [1] Philippe Moustrou, Cordian Riener, and Hugues Verdone. “Symmetries in polynomial optimization”. In: *Polynomial Optimization, Moments, and Applications*. Vol. 206. Springer Optimization and Its Applications. Springer, 2023, pp. 53–111.
- [2] Saugata Basu and Cordian Riener. “Efficient algorithms for computing the Euler-Poincaré characteristic of symmetric semi-algebraic sets”. In: *Ordered Algebraic Structures and Related Topics*. Vol. 697. Contemporary Mathematics. American Mathematical Society, 2017, pp. 53–81.

Books

- [1] Michal Kočvara, Bernard Mourrain, and Cordian Riener, eds. *Polynomial Optimization, Moments, and Applications*. Vol. 206. Springer Optimization and Its Applications. Springer, 2023.

Thesis

- [1] Cordian Benedikt Riener. “Symmetries in semidefinite and polynomial optimization: Relaxations, combinatorics, and the degree principle”. PhD thesis. Goethe-Universität Frankfurt, 2011.

Reports and other publications

- [1] Hans Z. Munthe-Kaas and Cordian Riener. “The Lie–Størmer Center”. In: *European Mathematical Society Magazine* 140 (2026), pp. 37–39.
- [2] Yoshio Okamoto, Cordian Riener, Markus Schweighofer, and Yoshiyuki Sekiguchi. “MFO-RIMS Tandem Workshop: Optimization, Theoretical Computer Science and Algebraic Geometry: Convexity and Beyond”. In: *Oberwolfach Reports* 22.1 (2025), pp. 405–450.
- [3] Gunnar Fløystad, Satoshi Murai, Cordian Riener, and Kohji Yanagawa. “MFO-RIMS Tandem Workshop: Symmetries on Polynomial Ideals and Varieties”. In: *Oberwolfach Reports* 18.3 (2022), pp. 2365–2396.